

Rajalakshmi Engineering College

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NeoColab_REC_CS23231_DATA STRUCTURES

REC_DS using C_Week 1_COD_Question 1

Attempt : 2
Total Mark : 10
Marks Obtained : 10

Section 1 : Coding

1. Problem Statement

Janani is a tech enthusiast who loves working with polynomials. She wants to create a program that can add polynomial coefficients and provide the sum of their coefficients.

The polynomials will be represented as a linked list, where each node of the linked list contains a coefficient and an exponent. The polynomial is represented in the standard form with descending order of exponents.

Input Format

The first line of input consists of an integer n , representing the number of terms in the first polynomial.

The following n lines of input consist of two integers each: the coefficient and the exponent of the term in the first polynomial.

The next line of input consists of an integer m, representing the number of terms in the second polynomial.

The following m lines of input consist of two integers each: the coefficient and the exponent of the term in the second polynomial.

Output Format

The output prints the sum of the coefficients of the polynomials.

Sample Test Case

Input: 3

2 2

3 1

4 0

3

2 2

3 1

4 0

Output: 18

Answer

```
#include <stdio.h>
```

```
#include <stdlib.h>
```

```
// Define a structure for the linked list node
```

```
typedef struct PolyNode {
```

```
    int coef;
```

```
    int exp;
```

```
    struct PolyNode* next;
```

```
} PolyNode;
```

```
// Function to create a new node
```

```
PolyNode* createNode(int coef, int exp) {
```

```
    PolyNode* newNode = (PolyNode*)malloc(sizeof(PolyNode));
```

```
    newNode->coef = coef;
```

```
    newNode->exp = exp;
```

```
    newNode->next = NULL;
```

```
    return newNode;
```

```
}
```

```
// Function to add a term to the polynomial linked list
```

```

void addTerm(PolyNode** head, int coef, int exp) {
    PolyNode* newNode = createNode(coef, exp);
    if (*head == NULL) {
        *head = newNode;
    } else {
        PolyNode* temp = *head;
        while (temp->next != NULL) {
            temp = temp->next;
        }
        temp->next = newNode;
    }
}

```

```

// Function to add two polynomials and sum the coefficients
int sumPolynomialCoefficients(PolyNode* poly1, PolyNode* poly2) {
    int sum = 0;
    PolyNode* p1 = poly1;
    PolyNode* p2 = poly2;

    while (p1 != NULL || p2 != NULL) {
        if (p1 == NULL) {
            sum += p2->coef;
            p2 = p2->next;
        } else if (p2 == NULL) {
            sum += p1->coef;
            p1 = p1->next;
        } else if (p1->exp == p2->exp) {
            sum += (p1->coef + p2->coef);
            p1 = p1->next;
            p2 = p2->next;
        } else if (p1->exp > p2->exp) {
            sum += p1->coef;
            p1 = p1->next;
        } else {
            sum += p2->coef;
            p2 = p2->next;
        }
    }
    return sum;
}

```

```

int main() {

```

```
int n, m, coef, exp;

// Read the first polynomial
PolyNode* poly1 = NULL;
scanf("%d", &n);
for (int i = 0; i < n; i++) {
    scanf("%d %d", &coef, &exp);
    addTerm(&poly1, coef, exp);
}

// Read the second polynomial
PolyNode* poly2 = NULL;
scanf("%d", &m);
for (int i = 0; i < m; i++) {
    scanf("%d %d", &coef, &exp);
    addTerm(&poly2, coef, exp);
}

// Calculate the sum of the coefficients
int result = sumPolynomialCoefficients(poly1, poly2);

// Output the result
printf("%d\n", result);

return 0;
}
```

Status : Correct

Marks : 10/10